OM RAJ

Software Developer • Environmental Analyst • Scientific Systems Engineer

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EXECUTIVE SUMMARY

Software developer, computational researcher, and environmental analyst with an **M.Sc. in Botany (Plant Pathology)** and project research experience at **IIT Ropar**. Specialized in building desktop software, web applications, and scientific calculation engines deployed on the **Microsoft Store** (*SNIFF DOI 2.0*, *WASH Suite v2.0*, *Calculator Luxe*, *Clock Span*). Expertly bridges biological research, environmental data processing (APHA 24th Ed / BIS 10500), and modern web/desktop engineering (JavaScript, HTML5, CSS3, Electron.js, Node.js, REST APIs).

RESEARCH, TECHNICAL & LABORATORY SKILLS

Analytical & Spectroscopy Instruments:
ICP-MS (Inductively Coupled Plasma Mass Spectrometry), UV-Visible Spectrophotometer, Gas Chromatography (GC), Ion Chromatography, TVOC Analyzer, Carbon Analyzer, Aerosol Spectrophotometer, Formaldehyde Analyzer.

Soil & Solid Waste Characterization:
Toxicity Characteristic Leaching Procedure (TCLP), acid digestion protocols, solid waste leachate extraction, heavy metal mobility profiling, bio-soil quality assessment, soil organic matter and nutrient testing.

Regulatory Auditing & Environmental Compliance:
Independent third-party monitoring, industrial compliance audits (CPCC, PPCB, Excise Dept.), wastewater testing (CETP, STP, ETP), hazardous waste leachate classification, and technical compliance reporting.

Software Development & Computing:
JavaScript, HTML5, CSS3, Electron.js, Node.js basics, REST API integration, Git/GitHub, Visual Studio Code.

Data Science, GIS & Automation:
MS Excel (advanced formulas, linear regression modeling), Power BI interactive dashboards, QGIS / basic ArcGIS, remote sensing interpretation.

Agriculture & Plant Pathology:
Crop disease diagnostics, pathogen identification, IDM, stress physiology, antioxidant enzyme assays (SOD, CAT, POD, proline, MDA), biochar amendments.

PROFESSIONAL RESEARCH & SOFTWARE EXPERIENCE

Project Researcher – Third-Party Environmental Monitoring & Soil Audit Project

Oct 2026 – Present

Dept. of Civil Engineering, IIT Ropar | Client/Sponsor: Chandigarh Pollution Control Committee (CPCC) | PI: Dr. Indramani Dhada

- **Field Sampling & Site Inspection:** Carried out field reconnaissance, verification, and composite soil sampling of legacy-waste bio-remediated material (bio-soil) excavated from Dadumajra Dumping Site across urban reclamation sites (Sector 47 & Sector 50, Chandigarh).
- **Sample Preparation & Heavy Metal Trace Analysis (ICP-MS):** Prepared solid matrices, carried out acid digestion/mineralization protocols, and performed trace elemental and heavy metal quantification using Inductively Coupled Plasma Mass Spectrometry (ICP-MS).
- **Leachability & Environmental Toxicity (TCLP):** Performed Toxicity Characteristic Leaching Procedure (TCLP) extractions to evaluate heavy metal mobility, leachate contamination hazards, and regulatory compliance per CPCB/hazardous waste guidelines.
- **Soil Quality & Risk Assessment:** Evaluated physicochemical parameters (pH, EC, organic content, reactivity), analyzed data, and contributed to compiling the independent third-party audit report.

Project Researcher & Scientific Systems Developer
2025 – Present

Department of Civil Engineering, IIT Ropar | Supervisors: Dr. Indramani Dhada, Dr. Neelkanth Nirmalkar | PM: Dr. Harsh Sharma

- **AWE LAB Academic Web Portal:** Architected, designed, and deployed the official research laboratory web portal for Dr. Indramani Dhada's research group at IIT Ropar.
- **Sustainable Agriculture & Biochar Project:** Evaluated comparative impact of agricultural biomass feedstocks (pineapple peel, jackfruit peel, drek wood biochars) and quantified defense enzyme kinetics (SOD, CAT, POD, Proline, MDA).
- **Wastewater & River Basin Monitoring (PPCB Project):** Conducted sampling and wet-chemical testing across CETPs, STPs, and Sutlej River basin for BOD, COD, TSS, heavy metals, and coliforms.
- **Technical Audits (Punjab Excise Dept.):** Participated in environmental audits of industrial distilleries, breweries, and bottling plants across Punjab; evaluated ZLD performance and compliance logs.

Postgraduate Dissertation Researcher — River Water Quality & Runoff
2024 – 2025

Patna University | Supervisor: Prof. Dr. Meenakshi Singh (Head, Dept. of Biotechnology)

- Executed multi-point seasonal field sampling along major ghats of Ganga River at Patna. Analyzed physicochemical and biological parameters against WHO, CPCB, and BIS 10500 standards.

Medical Writing Intern & Freelance Creative Specialist
2021 – Present

YuvaIntern (NSDC / Henry Harvin, Nov 2025) & Freelance / MARKAD LIMITED (2021–Present)

- Executed systematic literature reviews across PubMed, Scopus, Web of Science with standardized citations (APA, IEEE, Vancouver). Designed graphical abstracts and university election outreach media.

EDUCATION & ACADEMIC QUALIFICATIONS

Degree / Program	Specialization / Discipline	Institution / University	Year	Score / CGPA
M.Sc. in Botany	Plant Pathology	Department of Botany, Patna University	2025	**7.56 CGPA**
B.Sc. in Botany	Plant Sciences & Chemistry	Patna Science College, Patna University	2023	63.50%
Diploma in Computer Applications (DCA)	Software & Computing Systems	Vinita Computer Education	2022	64.40%
Senior Secondary (Class XII, Science)	Physics, Chemistry, Biology (PCB)	Shanti Niketan Sr. Sec. School (CBSE)	2019	58.40%
Secondary School (Class X)	General Science & Mathematics	Sunshine Prep. / High School (CBSE)	2017	81.70%

PUBLISHED SOFTWARE APPLICATIONS (MICROSOFT STORE)

- **AWE LAB Academic Portal (IIT Ropar):** Official research laboratory web portal for Dr. Indramani Dhada's research group at IIT Ropar. (AWE LAB Portal)
- **SNIFF DOI — Academic Literature Discovery Engine:** Published on Microsoft Store. Desktop application indexing 250M+ articles with 6-tier research gap analysis and CSL citation engine. (Microsoft Store)
- **WASH (Water Analysis Suite Hub):** Published on Microsoft Store. Compliance calculation engine for APHA 24th Ed & BIS 10500, spectrophotometer regression calibration, and NABL PDF export. (Microsoft Store)
- **Calculator Luxe & Clock Span (Watch Span):** High-precision scientific desktop tools live on Microsoft Store.

PUBLICATIONS & RESEARCH MANUSCRIPTS

- **Peer-Reviewed Journal Article:** Raj, O., & Ambreen, D. (2026). Assessment of Physicochemical Parameters of Water Samples from the Ganga River at Patna. *International Journal of Versatile Research and Analysis (IJVRA)*, 3(4), 784–810. ISSN: 2984-8903. Paper ID: IJVRA2603368.
- **Research Preprint (Air Quality):** Raj, O., & Dhada, I. Spatiotemporal Assessment of Ambient Air Quality in Selected Urban, Tourist and Industrial Regions of Himachal Pradesh, India: A Case Study. *Research Square*. DOI: 10.21203/rs.3.rs-10492149/v1
- **Manuscript in Preparation (Sustainable Agriculture):** Raj, O., & Dhada, I. Comparative Assessment of Pineapple Peel, Jackfruit Peel, and Drek Wood Biochar on Plant Growth, Soil Fertility, and Biochemical Responses. (Evaluates SOD, CAT, POD, proline, MDA kinetics).

WORKSHOPS & PROFESSIONAL CERTIFICATIONS

- **Fundamentals of Remote Sensing:** NASA "Train - Empower - Advance" Initiative (Sept 2026).
- **Advances in Genome Biology & Gene Regulation:** School of Life Sciences, JNU, New Delhi (June 2025).
- **Marine Data 4 Asia 2025:** Copernicus Marine Service & Mercator Ocean International, EU Program (June 2025).
- **Power BI & AI Interactive Dashboards:** OfficeMaster (Sept 2025).
- **Cancer Genomics, Clinical Research & AI:** Biotecnika (Sept 2025).

PROFESSIONAL REFERENCES

Dr. Indramani Dhada
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